

STATISTICS — MCQ ASSESSMENT

PART A — BASIC STATISTICS

1. What is the primary purpose of statistics?

- A. To store data
- B. To collect, analyze, interpret, and communicate data
- C. To increase the size of a dataset
- D. To convert all data into numbers

2. Which of the following is an example of quantitative data?

- A. Eye color
- B. Blood group
- C. Height
- D. Country

3. Which of the following is an example of categorical data?

- A. Age
- B. Weight
- C. Temperature
- D. Blood group

4. A population refers to:

- A. A small subset of observations
- B. The entire group being studied
- C. Only the observations used for training
- D. Only numerical observations

5. A sample is:

- A. The entire population
- B. A subset of the population
- C. Always larger than the population
- D. A single variable

6. Why is sampling commonly used?

- A. Studying the entire population may be expensive or impractical
- B. Samples always provide perfect results
- C. Populations cannot be measured
- D. Sampling eliminates all errors

PART B — MEASURES OF CENTRAL TENDENCY

7. Which measure represents the arithmetic average of a dataset?

- A. Median
- B. Mode
- C. Mean
- D. Range

8. What is the mean of:

10, 20, 30, 40, 50

- A. 20
- B. 25
- C. 30
- D. 35

9. The median is:

- A. The largest value
- B. The smallest value
- C. The middle value after arranging the data
- D. The most frequent value

10. What is the median of:

3, 7, 8, 10, 15

- A. 7
- B. 8
- C. 10
- D. 15

11. What is the median of:

2, 4, 6, 8

- A. 4
- B. 5
- C. 6
- D. 7

12. The mode represents:

- A. The average value
- B. The middle value
- C. The most frequently occurring value
- D. The largest value

13. Which dataset has a clear mode?

- A. 1, 2, 3, 4, 5
- B. 2, 2, 3, 4, 5
- C. 1, 2, 3, 4, 5, 6
- D. 10, 20, 30, 40

PART C — MEAN, MEDIAN AND OUTLIERS

14. Which measure is generally more sensitive to extreme values?

- A. Mean
- B. Median
- C. Mode
- D. Frequency

15. Consider:

10, 12, 13, 14, 100

Which measure would be most affected by the value 100?

- A. Mean
- B. Median
- C. Mode
- D. Minimum

16. If a very large outlier is added to a dataset, the mean will generally:

- A. Decrease
- B. Increase
- C. Remain exactly the same
- D. Become zero

17. For a highly skewed dataset containing extreme values, which measure may better represent the typical observation?

- A. Mean
- B. Median
- C. Variance
- D. Range

18. A dataset has mean = 50 and median = 50. What can we conclude?

- A. The distribution must be perfectly normal
- B. The distribution is necessarily symmetric
- C. Mean and median happen to be equal
- D. There are no outliers

PART D — MEASURES OF DISPERSION

19. What does dispersion describe?

- A. The center of the data
- B. How spread out the data are
- C. The number of observations
- D. The data type

20. Which of the following is a measure of dispersion?

- A. Mean
- B. Median
- C. Standard deviation
- D. Mode

21. The range of a dataset is:

- A. Mean – Median
- B. Maximum – Minimum
- C. Maximum + Minimum
- D. Median – Minimum

22. What is the range of:

5, 8, 10, 15, 20

- A. 10
- B. 15

- C. 20
- D. 25

23. If all values in a dataset are identical, the standard deviation is:

- A. 0**
- B. 1
- C. Equal to the mean
- D. Infinite

24. A larger standard deviation generally indicates:

- A. Less spread
- B. Greater spread**
- C. Smaller mean
- D. More observations

25. Two datasets have the same mean. Dataset A has standard deviation 2, while Dataset B has standard deviation 20. Which statement is most appropriate?

- A. Dataset A has greater variability
- B. Dataset B has greater variability**
- C. Both have identical variability
- D. Standard deviation cannot compare datasets

PART E — VARIANCE

26. Variance is based on:

- A. Absolute values from zero
- B. Squared deviations from the mean**
- C. The maximum value only
- D. The median

27. Why are deviations squared when calculating variance?

- A. To make all deviations positive and emphasize larger deviations**
- B. To make the mean zero
- C. To remove the observations
- D. To reduce the dataset size

28. If the variance of a dataset is 25, its standard deviation is:

- A. 5**
- B. 10
- C. 25
- D. 625

29. If the standard deviation is 4, the variance is:

- A. 2
- B. 4
- C. 8
- D. 16**

30. Which statement is correct?

- A. Variance and standard deviation always have the same units
- B. Standard deviation is the square root of variance
- C. Variance is the square root of standard deviation
- D. Variance is always smaller than standard deviation

PART F — DISTRIBUTIONS

31. A distribution describes:

- A. How values are arranged or occur in a dataset
- B. Only the largest value
- C. Only the average
- D. The number of columns

32. A symmetric distribution has:

- A. No mean
- B. Similar shape on both sides of its center
- C. Only positive values
- D. No variation

33. In a perfectly symmetric distribution, the mean and median are generally:

- A. Very different
- B. Approximately equal
- C. Always zero
- D. Undefined

34. A distribution with a long tail toward larger values is described as:

- A. Left-skewed
- B. Right-skewed
- C. Uniform
- D. Symmetric

35. A distribution with a long tail toward smaller values is:

- A. Right-skewed
- B. Left-skewed
- C. Uniform
- D. Normal

36. In a positively skewed distribution, which relationship is generally expected?

- A. Mean > Median
- B. Mean < Median
- C. Mean = 0
- D. Median = 0

PART G — NORMAL DISTRIBUTION

37. A normal distribution is generally:

- A. Symmetric and bell-shaped
- B. Always skewed

- C. Uniform
- D. Discrete only

38. In an ideal normal distribution, the mean, median, and mode are:

- A. Different
- B. Approximately equal
- C. Always zero
- D. Undefined

39. The standard deviation of a distribution determines:

- A. Its center only
- B. Its spread
- C. Its number of observations
- D. Its maximum value

40. Increasing the standard deviation of a normal distribution generally makes the distribution:

- A. Narrower
- B. Wider
- C. Shift to the right
- D. Shift to the left

41. Decreasing the standard deviation while keeping the mean unchanged makes a normal distribution:

- A. Wider
- B. Narrower
- C. More skewed
- D. Shifted right

PART H — STANDARDIZATION AND Z-SCORE

42. A z-score indicates:

- A. The original value of an observation
- B. How far an observation is from the mean in standard deviation units
- C. The number of observations
- D. The variance of the dataset

43. If an observation has a z-score of 0, it is:

- A. One standard deviation above the mean
- B. At the mean
- C. One standard deviation below the mean
- D. At the maximum

44. A z-score of +2 means that the observation is:

- A. 2 units above the mean
- B. 2 standard deviations above the mean
- C. 2 standard deviations below the mean
- D. 2 units below the mean

45. A z-score of -3 means that the observation is:

- A. 3 units below the mean
- B. 3 standard deviations below the mean
- C. 3 standard deviations above the mean
- D. Equal to the mean

PART I — CORRELATION

46. Correlation measures:

- A. The average of a variable
- B. The strength and direction of association between two variables
- C. The number of observations
- D. The variance of one variable only

47. A correlation coefficient close to +1 indicates:

- A. Strong positive linear association
- B. Strong negative linear association
- C. No linear association
- D. Causation

48. A correlation coefficient close to -1 indicates:

- A. Strong positive association
- B. Strong negative linear association
- C. No association
- D. Identical variables

49. A correlation coefficient close to 0 generally indicates:

- A. Strong positive linear relationship
- B. Strong negative linear relationship
- C. Little or no linear relationship
- D. Perfect causation

50. If height and weight have a positive correlation, it means:

- A. Increasing height necessarily causes weight to increase
- B. Taller individuals tend to have higher weight
- C. Every tall person is heavier
- D. Height and weight are identical variables

51. Which statement about correlation is correct?

- A. Correlation always implies causation
- B. Correlation does not necessarily imply causation
- C. Correlation can only be positive
- D. Correlation can only be used for categorical variables

PART J — QUANTILES AND PERCENTILES

52. The median divides an ordered dataset into:

- A. Two equal parts
- B. Three equal parts
- C. Four equal parts
- D. Ten equal parts

53. The first quartile (Q1) represents approximately:

- A. 10th percentile
- B. 25th percentile
- C. 50th percentile
- D. 75th percentile

54. The third quartile (Q3) represents approximately:

- A. 25th percentile
- B. 50th percentile
- C. 75th percentile
- D. 90th percentile

55. The interquartile range (IQR) is:

- A. $Q1 + Q3$
- B. $Q3 - Q1$
- C. $Q3 \times Q1$
- D. $Q1 / Q3$

56. Why is IQR useful?

- A. It measures the spread of the middle 50% of the data
- B. It measures the total number of observations
- C. It always equals the standard deviation
- D. It identifies the mean

PART K — SAMPLING

57. Random sampling means:

- A. Selecting only the largest observations
- B. Giving members of the population a chance of being selected
- C. Selecting observations manually
- D. Selecting only convenient observations

58. A sample that does not properly represent the population may lead to:

- A. Sampling bias
- B. Zero variance
- C. Perfect prediction
- D. Normal distribution

59. Increasing the sample size generally:

- A. Makes estimates less reliable
- B. Can make estimates more stable
- C. Eliminates all sources of bias
- D. Guarantees a normal population

60. Which of the following is most likely to produce a biased sample?

- A. Randomly selecting people from the entire population
- B. Selecting participants only from one convenient location

- C. Using a properly designed random sampling method
- D. Repeating random sampling several times

PART L — INTERPRETATION

61. A dataset has:

Mean = 50

Median = 50

Standard deviation = 2

Which statement is most reasonable?

- A. The values are generally close to 50
- B. The values are extremely spread out
- C. The mean must be an outlier
- D. There is no variation

62. A dataset has:

Mean = 50

Standard deviation = 30

What does this suggest?

- A. The data have relatively high variability around the mean
- B. All values are close to 50
- C. All observations equal 50
- D. The dataset has no variance

63. Two datasets have identical means and different standard deviations. What does this tell us?

- A. They have exactly the same distribution
- B. Their centers are the same, but their variability differs
- C. Their medians must be different
- D. Their sample sizes must be different

64. A dataset has a strong positive correlation between two variables. Which conclusion is safest?

- A. One variable definitely causes the other
- B. The variables have a strong positive linear association
- C. The variables must have identical values
- D. There is no relationship

65. Which statistic is generally most appropriate for describing the center of highly skewed data with extreme outliers?

- A. Mean
- B. Median
- C. Variance
- D. Standard deviation